

Research Notes on RoBERTa

RoBERTa

>> Section 1

Bidirectional encoder representations from transformers (BERT) is a language model introduced in October 2018 by researchers at Google. It learns to represent text as a sequence of vectors using self-supervised learning. It uses the encoder-only transformer architecture. BERT dramatically improved the state of the art for large language models. As of 2020, BERT is a ubiquitous baseline in natural language processing (NLP) experiments. BERT is trained by masked token prediction and next sentence prediction. With this training, BERT learns contextual, latent representations of tokens in their context, similar to ELMo and GPT-2. It found applications for many natural language processing tasks, such as coreference resolution and polysemy resolution. It improved on ELMo and spawned the study of "BERTology", which attempts to interpret what is learned by BERT. BERT was originally implemented in the English language at two model sizes, BERTBASE (110 million parameters) and BERTLARGE (340 million parameters). Both were trained on the Toronto BookCorpus (800M words) and English Wikipedia (2,500M words). The weights were released on GitHub. On March 11, 2020, 24 smaller models were released, the smallest being BERTTINY with just 4 million parameters.

>> Section 2

BERT is meant as a general pretrained model for various applications in natural language processing. That is, after pre-training, BERT can be fine-tuned with fewer resources on smaller datasets to optimize its performance on specific tasks such as natural language inference and text classification, and sequence-to-sequence-based language generation tasks such as question answering and conversational response generation. The original BERT paper published results demonstrating that a small amount of finetuning (for BERTLARGE, 1 hour on 1 Cloud TPU) allowed it to achieve state-of-the-art performance on a number of natural language understanding tasks:

>> Section 3

History BERT was originally published by Google researchers Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. The design has its origins from pre-training contextual representations, including semi-supervised sequence learning, generative pre-training, ELMo, and ULMFit. Unlike previous models, BERT is a deeply bidirectional, unsupervised language representation, pre-trained using only a plain text corpus. Context-free models such as word2vec or GloVe generate a single word embedding representation for each word in the vocabulary, whereas BERT takes into account the

context for each occurrence of a given word. For instance, whereas the vector for "running" will have the same word2vec vector representation for both of its occurrences in the sentences "He is running a company" and "He is running a marathon", BERT will provide a contextualized embedding that will be different according to the sentence. On October 25, 2019, Google announced that they had started applying BERT models to English-language search queries on Google Search within the US. On December 9, 2019, it was reported that BERT had been adopted by Google Search for over 70 languages. In October 2020, almost every single English-based query was processed by a BERT model.

>> Section 4

GLUE (General Language Understanding Evaluation) task set (consisting of 9 tasks); SQuAD (Stanford Question Answering Dataset) v1.1 and v2.0; SWAG (Situations With Adversarial Generations). In the original paper, all parameters of BERT are fine-tuned, and recommended that, for downstream applications that are text classifications, the output token at the [CLS] input token is fed into a linear-softmax layer to produce the label outputs. The original code base defined the final linear layer as a "pooler layer", in analogy with global pooling in computer vision, even though it simply discards all output tokens except the one corresponding to [CLS] .

>> Section 5

In masked language modeling, 15% of tokens would be randomly selected for masked-prediction task, and the training objective was to predict the masked token given its context. In more detail, the selected token is:

>> Section 6

Given two sentences, the model predicts if they appear sequentially in the training corpus, outputting either [IsNext] or [NotNext]. During training, the algorithm sometimes samples two sentences from a single continuous span in the training corpus, while at other times, it samples two sentences from two discontinuous spans. The first sentence starts with a special token, [CLS] (for "classify"). The two sentences are separated by another special token, [SEP] (for "separate"). After processing the two sentences, the final vector for the [CLS] token is passed to a linear layer for binary classification into [IsNext] and [NotNext]. For example:

>> Section 7

Interpretation Language models like ELMo, GPT-2, and BERT, spawned the study of "BERTology", which attempts to interpret what is learned by these models. Their performance on these natural language understanding

Research Notes on RoBERTa

RoBERTa

tasks are not yet well understood. Several research publications in 2018 and 2019 focused on investigating the relationship behind BERT's output as a result of carefully chosen input sequences, analysis of internal vector representations through probing classifiers, and the relationships represented by attention weights. The high performance of the BERT model could also be attributed to the fact that it is bidirectionally trained. This means that BERT, based on the Transformer model architecture, applies its self-attention mechanism to learn information from a text from the left and right side during training, and consequently gains a deep understanding of the context. For example, the word fine can have two different meanings depending on the context (I feel fine today, She has fine blond hair). BERT considers the words surrounding the target word fine from the left and right side. However it comes at a cost: due to encoder-only architecture lacking a decoder, BERT can't be prompted and can't generate text, while bidirectional models in general do not work effectively without the right side, thus being difficult to prompt. As an illustrative example, if one wishes to use BERT to continue a sentence fragment "Today, I went to", then naively one would mask out all the tokens as "Today, I went to [MASK] [MASK] [MASK] ... [MASK] ." where the number of [MASK] is the length of the sentence one wishes to extend to. However, this constitutes a dataset shift, as during training, BERT has never seen sentences with that many tokens masked out. Consequently, its performance degrades. More sophisticated techniques allow text generation, but at a high computational cost.

>> Section 8

Cost BERT was trained on the BookCorpus (800M words) and a filtered version of English Wikipedia (2,500M words) without lists, tables, and headers. Training BERTBASE on 4 cloud TPU (16 TPU chips total) took 4 days, at an estimated cost of 500 USD. Training BERTLARGE on 16 cloud TPU (64 TPU chips total) took 4 days.